朴素贝叶斯
Naïve Bayes

明玉瑞 Yurui Ming

yrming@gmail.com

声明
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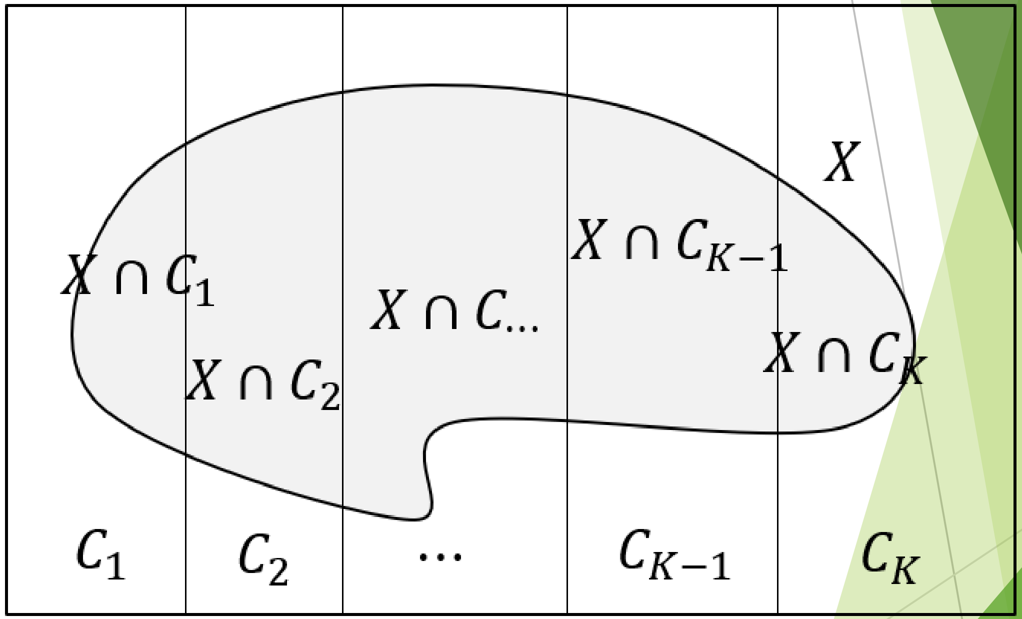
贝叶斯定理
Bayes Rule

- 假设$\{C_1, C_2, \cdots, C_K\}$为$S$的一个划分，即对$i \neq j$，$C_i \neq C_j$；若对$k = 1,2,\cdots,K$，$\mathrm{P}(C_k) > 0$，则有：

  Assume that $\{C_1, C_2, \cdots, C_K\}$ is a partition of S such that $\mathrm{P}(C_k) > 0$, for $k = 1,2,\cdots,K$, Then:

$$\mathrm{P}(C_i|X) = \frac{\mathrm{P}(C_i, X)}{\mathrm{P}(X)} = \frac{\mathrm{P}(C_i, X)}{\sum_{k=1}^{K} \mathrm{P}(C_k, X)}$$

$$= \frac{\mathrm{P}(X|C_i)\mathrm{P}(C_i)}{\sum_{k=1}^{K} \mathrm{P}(X|C_k)\mathrm{P}(C_k)}$$



贝叶斯定理应用
Application of Bayes Rule

- 我们考虑信用卡评级的例子：根据过去的交易记录，一些客户属于低风险，因为他们能够按时偿还贷款，银行也从他们身上获得了利润；而另一些客户则属于高风险，因为他们出现了违约。

We consider the credit cards scoring example: According to the past transactions, some customers are low-risk in that they paid back their loan and the bank profited from them and other customers are high-risk in that they defaulted.

- 银行往往希望知道潜在的高风险客户群。我们用两个随机变量X_1和X_2分别表示客户的年收入和储蓄，客户的信用状况表示为伯努利随机变量C表示，其中$C = 1$表示高风险客户，$C = 0$表示低风险客户。

The banks always want to identify the high-risk customer group. We represent customer's yearly income and savings by two random variables X_1 and X_2, respectively. The credibility of a customer is denoted by a Bernoulli random variable C where $C = 1$ indicates a high-risk customer and $C = 0$ indicated a low-risk customer

贝叶斯定理应用

Application of Bayes Rule

- 实际上，X_1和X_2的值依赖于用户本身的诚信度，由于文化与传统的原因，不同地区的人们对诚信度的价值观可能不同。记$c_0 \triangleq \{C=1\}$，$c_1 \triangleq \{C=1\}$，对某个地区而言，$\mathrm{P}(c_0) > \mathrm{P}(c_1)$，则收到一份信用卡申请时，我们是应该拒绝还是接受呢？

Actually, the values of X_1 and X_2 depend on the user's own sense of honesty. Due to cultural and traditional reasons, people in different regions may have different values regarding honesty. Let $c_0 \triangleq \{C=1\}$ and $c_1 \triangleq \{C=1\}$. For a certain region, if $\mathrm{P}(c_0) > \mathrm{P}(c_1)$, then when we receive a credit card application, should we reject or accept it?

- 根据经验，若$\mathrm{P}(c_0) > \mathrm{P}(c_1)$，我们倾向于接受申请。实际上，由于$\mathrm{P}(c_0) + \mathrm{P}(c_1) = 1$，根据上面规则，我们所犯错误的概率为我们所犯错误的概率为：$1 - \max\{\mathrm{P}(c_0), \mathrm{P}(c_1)\} < 0.5$。

Based on experience, if $\mathrm{P}(c_0) > \mathrm{P}(c_1)$, we tend to accept the application. In fact, since $\mathrm{P}(c_0) + \mathrm{P}(c_1) = 1$, according to the above rule, the probability of making a mistake is $1 - \max\{\mathrm{P}(c_0), \mathrm{P}(c_1)\} < 0.5$.

贝叶斯定理应用
Application of Bayes Rule

- 接受或拒绝一个客户申领信用卡，并不会产生严重的后果，但如果是其他情况的分类呢？比如对某种疾病的阳性或阴性？因此要把更多的因素纳入考量。

Accepting or rejecting a customer's credit card application does not have serious consequences, but what about classification in other situations? For example, determining whether a disease is positive or negative? Therefore, more factors need to be taken into consideration.

- 假设我们收到一份信用卡申请，特征为$X_1 = x_1$和$X_2 = x_2$，额外的，如果我们知道在给定观测$x \triangleq X = [x_1; x_2]$条件下$C$的概率，我们应该怎么做呢？

Suppose we receive a credit card application with features $X_1 = x_1$and $X_2 = x_2$. Additionally, if we know the probability of C given the observed $X = [x_1; x_2]$, what should we do?

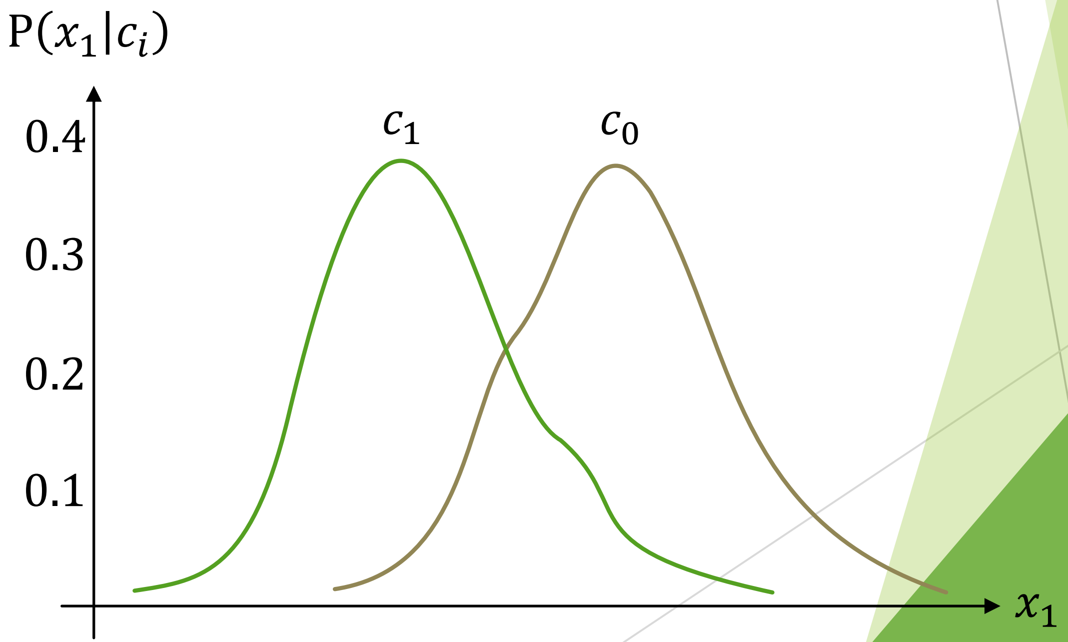
贝叶斯定理应用
Application of Bayes Rule

- 假设根据既往数据，我们知道年收入与不同风险客户群体的关系如下图所示：

  Suppose that based on past data, we know the relationship between annual income and different risk customer groups, as shown in the figure below:

- 由于我们有了额外的信息，即$P(x_1|c_i)$，那么如果有包含年收入的新的信用卡申请，则我们单纯地由直接考虑$P(C)$，转向考虑概率$P(C|x_1)$。

  Since we now have additional information, namely $P(x_1|c_i)$, when a new credit card application arrives that includes the annual income, we shift from simply considering $P(C)$ to considering the probability $P(C|x_1)$.



贝叶斯定理应用
Application of Bayes Rule

- 概率$\mathrm{P}(C|x) = \frac{\mathrm{P}(x|C)\mathrm{P}(C)}{\mathrm{P}(x)}$称为后验概率。
- $\mathrm{P}(C=1)$被称为$C=1$的先验概率。在我们的例子中，它对应于不考虑x的情况下，一个客户是高风险的概率。之所以称为先验概率，是因为它是我们在观察到x之前所拥有的知识。
- $\mathrm{P}(x|C)$被称为类条件似然，是属于类别C的事件产生观测值x的条件概率。
- $\mathrm{P}(x)$是证据，是在不考虑它是正例还是反例的情况下，观测到x的概率。
- 以上所有信息都可以从过去的交易记录（历史数据）中提取。

- The probability $\mathrm{P}(C|x) = \frac{\mathrm{P}(x|C)\mathrm{P}(C)}{\mathrm{P}(x)}$ is called posterior probability.
- $\mathrm{P}(C=1)$ is called the prior probability that $C=1$. In our example, it corresponds to a probability that a customer is high-risk, regardless of the x value. It is called the prior probability because it is the knowledge we have before looking at the observation x.
- $\mathrm{P}(x|C)$ is called the class likelihood and is the conditional probability that an event belonging to the class C has the associated observation value x.
- $\mathrm{P}(x)$, the evidence is the probability that an observation x to be seen, regardless of whether it is a positive or negative example.
- All above information can be extracted from the past transactions (historical data).

贝叶斯定理应用
Application of Bayes Rule

- 那么，当我们收到一份信用卡申请时，我们应该怎么做呢？我们通过观测特征X_1的取值，将先验概率$\mathrm{P}(C)$转换为后验概率$\mathrm{P}(C|x_1)$——即在已测到特征值$X_1 = x_1$的情况下，该样本属于高风险类别的概率。

  How to make the decision when a new application arrives? Convert the prior probability $\mathrm{P}(C)$ to the a posteriori probability (or posterior) probability $\mathrm{P}(C|x_1)$ − the probability of the high risk instance being x_1 given that feature value X_1 has been measured − by observing the value of X_1.

- 从贝叶斯观点看，后验概率比先验更可靠、更有说服力，因为它用观测到的证据（数据）去更新最初的先验信念。先验只反映“看到数据之前”的判断，而后验把先验与新证据结合起来，因此更贴近实际情况。

  From a Bayesian perspective, the posterior probability is more reliable and persuasive than the prior because it updates the initial belief (prior) with actual observed evidence (data). The prior only reflects what we knew before seeing the data, while the posterior combines prior knowledge with new information, making it better aligned with reality.

贝叶斯定理应用
Application of Bayes Rule

- 当我们收到一份特征为$X_1=x_1$和$X_2=x_2$的信用卡申请时，我们应该怎么做呢？如果我们计算出在给定观测$x \triangleq X=[x_1;x_2]$条件下C的概率，那么我们的决策规则为：若$\mathrm{P}(C=1|x)>0.5$，则判定$C=1$否则判定$C=0$。

  How to make the decision when a new application arrives with $X_1=x_1$ and $X_2=x_2$. If we know the probability of C conditioned on the observation $X=[x_1;x_2]$, our decision will be $C=1$ if $\mathrm{P}(C=1|x)>0.5$, $C=0$ otherwise.

- 实际上，由于通过证据进行了归一化，后验概率$\mathrm{P}(C=1|x)+\mathrm{P}(C=0|x)=1$。基于上面决策规则，我们所犯错误的概率为：$1-\max\{\mathrm{P}(C=1|x),\mathrm{P}(C=0|x)\}<0.5$。该规则称为贝叶斯决策理论。

  Notably, because of normalization by the evidence, the posteriors $\mathrm{P}(C=1|x)+\mathrm{P}(C=0|x)=1$. The probability of error we made based on this rule is $1-\max\{\mathrm{P}(C=1|x),\mathrm{P}(C=0|x)\}<0.5$. This rule is called Bayesian decision theory.

贝叶斯定理应用
Application of Bayes Rule

- 在假设这两个随机变量X_1和X_2相互独立的前提下，根据联合概率的性质，我们有$\mathrm{P}(X_1, X_2) = \mathrm{P}(X_1)\mathrm{P}(X_2)$，
- 尽管根据抽样实践，我们对抽样中涉及的属性尽可能保持其正交，但这仍是对问题进行了一定的简化。
- 上述假设是这类贝叶斯分类器称为朴素的原因之一，不过，实践证明，该假设还是非常方便和有用的。

- Under the assumption, these two random variables X_1 and X_2 are probability independent, according to the property of joint probability, we have $\mathrm{P}(X_1, X_2) = \mathrm{P}(X_1)\mathrm{P}(X_2)$.
- Although by adopting the practice of sampling, we try to make sure the properties being considered are perpendicular with each other, it is still an simplified version of the problem.
- It is one of key reasons for calling this Bayes' Classifier being naive. However, it is very convenient and useful for real applications.

贝叶斯定理应用
Application of Bayes Rule

- 假设我们有K个互斥且穷尽的类别：C_i，其中$i = 1, 2, \cdots, K$。
- 例如，在光学数字识别中，输入是位图图像，共有 10 个类别。我们可以将这K个类别视为对输入空间的一个划分。
- 贝叶斯分类器会选择后验概率最高的类别；也就是说，若$\mathrm{P}(C_i|x) = \max_k\{\mathrm{P}(C_k|x)\}$，我们就选择类别$C_i$。
- 问题：$\mathrm{P}(x)$（即证据）是否非常重要？

- We have K mutually and exhaustive classes: C_i, $i = 1, 2, \cdots, K$.
- For example, in optical digit recognition, the input is a bitmap image and there are 10 classes. We can think of that these K classes define a partition of the input space.
- The Bayes‘ classifier choose the class with the highest posterior probability; that is we choose C_i if $\mathrm{P}(C_i|x) = \max_k\{\mathrm{P}(C_k|x)\}$.
- Question: Is it very important to have $\mathrm{P}(x)$, the evidence?